

Assignment #2**Name: Abdullah Gamal Moustafa Elafifi****ID: 20221238****Summarized Report:**

For this assignment, I implemented the Genetic Algorithm (GA) to solve the optimization problem:

$$\text{Maximize } F(x_1, x_2) = 8 - (x_1 + 0.0317)^2 + x_2^2, \text{ where } -2 \leq x_1, x_2 \leq 2.$$

The GA starts with an initial population of randomly generated binary chromosomes representing x_1 and x_2 . Two types of decoding methods were explored: standard binary decoding and Gray code decoding. The encoded values were then mapped to real values within the given range. The fitness function was evaluated using the problem's objective function.

The selection of parents was based on a probability proportional to their fitness scores using roulette wheel selection. A single-point crossover was applied with a given probability to generate offspring. Bit-flip mutation was then applied to introduce genetic diversity. Elitism was used to ensure that the best-performing individuals were carried over to the next generation.

In Part II, an additional constraint was introduced: $x_1 + x_2 = 1$. The fitness function was adjusted to incorporate a penalty term, penalizing individuals that violated this constraint.

Experimental Desing (as requested in the assignment):

- Population Size: **20**
- Chromosome Length: **5**
- Generations: **100**
- Crossover Probability: **0.6**
- Mutation Probability: **0.05**
- Decoding Methods: **Standard binary and Gray decoding (5, 8, and 12)**
- Additional constraint: **$x_1 + x_2 = 1$**
- Fitness function adjusted: **$F(x_1, x_2) - |x_1 + x_2 - 1|$**

The Best Fitness Values:

Was calculated per generation were recorded, to get a snapshot of improvement in fitness over time

Tools and Technologies:

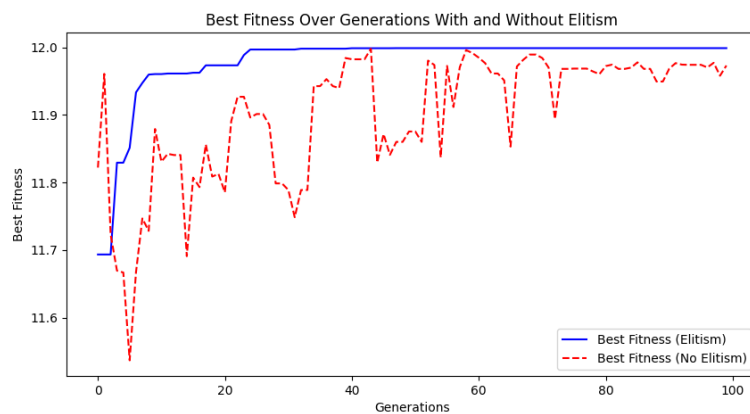
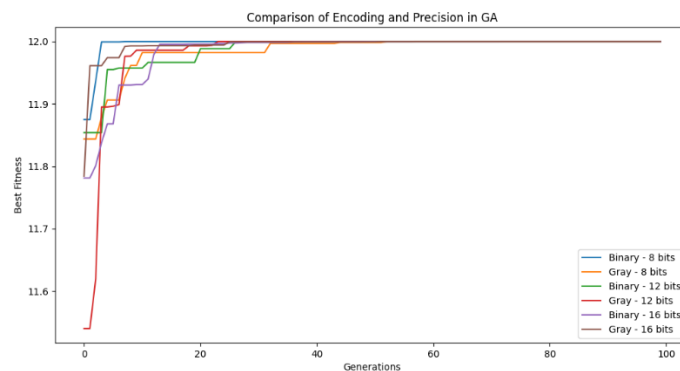
- Python
- Numpy Library
- Random Library
- Matplotlib for visualization

I have put a random seed to ensure the reproducibility of the code per each run:

`np.random.seed( X )` (*manually*)

These are the results of the 10 time of manual run:

As well as the “Average Fitness per Generation” and “Best Fitness per Generation” but here will be a very long code view.



Final Results (with Seed 2, and example):

- With Elitism - Best Fitness: 11.998963906873998
- With Elitism - Average Fitness: 10.765238895850352
- Without Elitism - Best Fitness: 11.99806381733391
- Without Elitism - Average Fitness: 10.513666819095931

Conclusion and Analysis:

The experimental results showed that the population evolved towards higher fitness individuals. Initially, the chromosomes exhibited a broad range of diversity. As generations progressed, selection, crossover, and mutation operations refined the population towards better solutions. Gray code often resulted in smoother convergence, as small mutations led to smaller changes in real values. Moreover, Increasing the number of bits for encoding improved the accuracy of the solution but also increased computational complexity.

The use of elitism helped maintain the best solutions across generations, leading to faster convergence, and adding the constraint $x_1 + x_2 = 1$ affected the fitness landscape, and solutions had to balance between maximizing the function and satisfying the constraint.

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